CSE437 — Data Science

Project Milestone 1

Literature Review

Project Title:

Customer Segmentation for Targeted Marketing Using Machine Learning

Dataset: Customer Personality Analysis — Kaggle

https://www.kaggle.com/datasets/imakash3011/customer-personality-analysis

Group Members

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Milestone 1

Each member must review at least 5 research papers. Share all papers via a Google Drive link. Then in this Google Doc, each member adds their own literature review with two sections:

Section 1 — Summarize each paper individually in 150–200 words.

Section 2 — Combine all summaries into a single cohesive narrative of 600–1000 words, written in a storytelling style, possibly spanning more than one paragraph.

Google Drive Link (Papers, BibTeX): TRISHA BibTeX LINK ,

Riajul Islam

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Section 1 — Individual Paper Summaries (5 Papers Required, 150–200 words each)

📌 *Instructions: Summarize each paper in 150–200 words. For each paper, include: the research problem, methodology, findings, and relevance to our project topic.*

Paper 1: A Mathematical Model for Customer Segmentation Leveraging Deep Learning, Explainable AI, and RFM Analysis in Targeted Marketing

> This paper addresses the critical limitation of traditional customer segmentation methods, which often fail to capture complex, non-linear patterns in large-scale marketing datasets. The authors introduce DeepLimeSeg, a novel framework that integrates deep learning (DL) with LIME-based Explainable AI (XAI) to segment customers into meaningful, interpretable groups. The methodology combines RFM (Recency, Frequency, Monetary) analysis as a feature engineering step with K-means clustering and a deep neural network architecture. LIME explanations are then applied to make the model's decisions transparent and actionable for marketing practitioners. Experiments were conducted on both the Mall Customer Segmentation dataset and an E-Commerce dataset from Kaggle. The model achieved an MSE of 0.9412 and an MAE of 0.9874, outperforming baseline approaches such as standard K-means and SVM-based segmentation. The paper is directly relevant to our project because it demonstrates how combining classical RFM analysis with modern deep learning can produce richer, more accurate customer segments. It also highlights the growing need for explainability in AI-driven marketing systems, a gap our project can further explore using the Customer Personality Analysis dataset.

Paper 2: Machine Learning-Driven Customer Segmentation and Targeted Marketing: A Systematic Review

> This systematic review comprehensively surveys the landscape of machine learning techniques applied to customer segmentation and targeted marketing across multiple industries. The authors adopted the PRISMA methodology, screening publications based on relevance, citation count, accuracy metrics, and recency. The review covers a wide spectrum of algorithms including K-means, DBSCAN, XGBoost, Self-Organizing Maps (SOM), and deep learning-based approaches, analyzing their strengths and limitations in the segmentation context. Key findings reveal that while K-means remains the most widely used algorithm due to its simplicity, ensemble methods such as XGBoost and hybrid deep learning models consistently achieve superior performance on complex, high-dimensional datasets. The review also addresses critical challenges including data quality issues, algorithm selection trade-offs, and ethical concerns around customer data usage. For our project, this paper serves as a comprehensive map of the existing research landscape, helping identify which algorithms are most promising for the Customer Personality Analysis dataset. It also highlights open research gaps — particularly around combining dimensionality reduction techniques like PCA with clustering — that our project can directly build upon.

Paper 3: Customer Personality Analysis and Clustering for Targeted Marketing

> This paper directly tackles the same research problem as our project, making it one of the most relevant studies in our literature review. The authors argue that a customer's personality is a key determinant of purchasing behavior, and therefore personality-based segmentation produces more actionable marketing insights than purely demographic or transactional approaches. The study applies the K-means clustering algorithm to a customer personality dataset, determining the optimal number of clusters using the Elbow method. Feature engineering incorporates demographic attributes, purchase histories, and behavioral indicators such as product category preferences and campaign responses. Results reveal distinct personality-driven customer segments with different

spending patterns and channel preferences, enabling targeted marketing strategies tailored to each group. The paper also evaluates the role of dimensionality reduction techniques such as PCA in improving cluster quality when dealing with high-dimensional personality data. This study is critical to our project because it validates the use of the Customer Personality Analysis Kaggle dataset for segmentation tasks, provides a methodological baseline our team can replicate and extend, and demonstrates the real-world marketing value of personality-based customer clustering.

Paper 4: [Predictive Modeling of Consumer Purchase Behavior on Social Media: Integrating Theory of Planned Behavior and Machine Learning for Actionable Insights]

This paper presents a unique interdisciplinary approach that merges the Theory of Planned Behavior (TPB) — a well-established psychological framework — with machine learning models to predict consumer purchase behavior on social media platforms. Conducted by researchers at North South University, Dhaka, Bangladesh, the study examines three core TPB factors: attitude toward buying, subjective social norms, and perceived behavioral control. The authors collected survey data from social media users and applied multiple ML classifiers including Logistic Regression, Random Forest, SVM, and Decision Trees, comparing their predictive accuracy. Random Forest and SVM demonstrated the highest accuracy in classifying purchase intent. The study finds that social norms and attitude are the strongest predictors of purchase behavior, with machine learning significantly improving prediction accuracy over traditional statistical models. For our project, this paper is especially relevant for two reasons: first, it demonstrates how personality and behavioral attributes — similar to those in our dataset — can predict purchase outcomes; second, it was conducted in Bangladesh, directly reflecting the consumer behavior context most relevant to our team and providing a culturally relatable benchmark.

Paper 5: [Application of Machine Learning in Predicting Consumer Behavior and Precision Marketing]

This 2025 study investigates the practical application of four advanced machine learning models — Support Vector Machine (SVM), XGBoost, CatBoost, and Backpropagation Artificial Neural Network (BPANN) — for predicting consumer purchase intention and enabling precision marketing strategies in e-commerce environments. Using a large-scale e-commerce transactional dataset, the authors perform rigorous data preprocessing including feature engineering, normalization, and handling of class imbalance. Among all models tested, CatBoost achieved the highest performance with an F1 score of 0.93 and ROC AUC of 0.985, followed closely by XGBoost with an F1 score of 0.92. The paper further demonstrates how these predictive models can directly inform personalized recommendation systems, dynamic pricing strategies, and precision advertising campaigns. Critically, the authors find that combining customer segmentation (to identify high-value groups) with purchase intent prediction (to time interventions correctly) produces the strongest marketing outcomes. This paper is highly valuable for our project as it benchmarks state-of-the-art ML models on consumer behavior data, provides concrete evaluation metrics we can replicate, and introduces the concept of segment-level prediction — a natural extension of the Customer Personality Analysis segmentation task our team is pursuing.

Trisha Das
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Paper 1: [K-Means Clustering Approach for Intelligent Customer Segmentation Using Customer Purchase Behavior Data]

Citation: *Tabianan, K., Velu, S., & Ravi, V. (2022). Sustainability, 14(12), 7243.*

The study by Tabianan et al. addresses the problem of customer churn in e-commerce platforms and argue that displaying irrelevant or overpriced products to customers due to poor segmentation leads to high churn rates and lost revenue. The authors employed K-Means clustering alongside data mining techniques and a dashboard based visualization system to segment customers based on their purchase behavior. The research demonstrates that K-Means clustering, when applied to customer purchase data, allows businesses to identify distinct customer groups and determine how to deal with each category in order to maximize profit. The dataset used was drawn from an e-commerce platform and the Elbow method was used to determine the optimal number of clusters. Findings showed that segmentation enabled businesses to tailor their offerings based on distinct behavioral profiles. The paper also emphasized the value of visual dashboards for communicating segment insights to business stakeholders. This study is directly relevant to our project as it applies K-Means clustering to behavioral purchase data, a methodology we replicate on the Customer Personality Analysis dataset, which similarly includes product spending and purchase behavior features across multiple categories.

Paper 2: [An Exploration of Clustering Algorithms for Customer Segmentation in the UK Retail Market]

Citation: *John, J. M., Shobayo, O., & Ogunleye, B. (2023). Analytics, 2(4), 809–823.*

The study by John et al. addresses the need for sophisticated approaches to perform more accurate and efficient customer segmentation amid growing volumes of online retail transactions, framing customer segmentation as a marketing analytical tool that aids customer-centric service and enhances profitability. The authors employed a UK-based online retail dataset from the UCI machine learning repository containing 541,909 customer records and eight features, adopting the RFM framework to quantify customer values. They then compared several clustering algorithms: K-Means, Gaussian Mixture Model (GMM), DBSCAN, Agglomerative Clustering and BIRCH. Results showed that GMM outperformed all other approaches, achieving a Silhouette Score of 0.80. This comparative study is highly relevant to our project because it provides a rigorous benchmarking framework for choosing between clustering algorithms, a decision we face on the Customer Personality Analysis dataset. The RFM-based preprocessing and the evaluation metrics named Silhouette Score used here directly mirror what we plan to employ in validating our own segmentation results.

Paper 3: How Can Algorithms Help in Segmenting Users and Customers? A Systematic Review and Research Agenda for Algorithmic Customer Segmentation

Citation: *Salminen, J., Mustak, M., Sufyan, M., & Jansen, B. J. (2023). Journal of Marketing Analytics, 11(4), 677–692.*

The study by Salminen et al. conducts a systematic literature review to address central practical questions in customer segmentation research such as which algorithm to choose, whether to use one or many, how many segments to create and how to evaluate results. By extracting information from 172 relevant articles, the authors found that algorithmic customer segmentation is the predominant approach with researchers employing 46 different algorithms and 14 different evaluation metrics. K-Means clustering emerged as the most frequently used algorithm. The study also found that separation focused metrics are slightly more prevalent than statistics-focused ones and that extant studies rarely use domain experts in evaluating segmentation outcomes. This paper is particularly valuable for our project as it provides a comprehensive methodological landscape for justifying our algorithm selection. Its finding that K-Means is the dominant approach and its critique of relying solely on statistical metrics reinforce the need to complement our cluster evaluation with business-level interpretation when analyzing the Customer Personality Analysis dataset.

Paper 4: Customer Profiling, Segmentation, and Sales Prediction Using AI in Direct Marketing

Citation: *Kasem, M. S., Hamada, M., & Taj-Eddin, I. (2024). Neural Computing and Applications, 36(9), 4995–5005.*

The study by Kasem et al. investigates customer clustering using RFM analysis combined with validation metrics to derive optimal segments with the overarching goal of improving sales performance and customer profiling for digital start-ups in direct marketing contexts. The K-Means clustering algorithm was applied alongside the Elbow method, Silhouette coefficient and Gap Statistics method to identify three primary clusters: new customers, best customers and intermittent customers. For each cluster, distinct strategies were proposed such as tailored learning content for new customers and personalized incentives for best customers demonstrating the direct link between segmentation and actionable marketing decisions. The paper also incorporates a boosting tree model for sales prediction on top of the segmentation results. This is highly relevant to our project because it not only applies RFM-based K-Means clustering techniques mirrored in our use of the Customer Personality Analysis dataset but also demonstrates how segmentation outputs can be extended into predictive models for campaign response, closely mirroring our project's downstream goals.

Paper 5: Customer Segmentation in Digital Marketing Using a Q-Learning Based Differential Evolution Algorithm Integrated with K-Means Clustering

Citation: *Wang, G. (2025). PLOS ONE, 20(2), e0318519.*

The study by Wang addresses the challenges of customer segmentation in digital marketing, proposing a novel framework called K-means-QLDE that integrates a reinforcement learning-based differential evolution algorithm with K-Means clustering using dimensionality reduction techniques. The methodology first employs Principal Component Analysis (PCA) to eliminate redundant and multicollinear features and then introduces a Q-learning-based parameter adaptive adjustment to overcome the fixed scaling factor limitation of traditional differential evolution algorithms. Using a customer dataset sourced from Kaggle, the method segments customers into six distinct clusters representing varying levels of importance, validated using four widely recognized classification accuracy methods. The study is directly applicable to our project in two ways: it uses a Kaggle-sourced customer dataset similar to ours and its combination of PCA for dimensionality reduction with K-Means for clustering mirrors our core pipeline. The paper also demonstrates how reinforcement learning can adaptively optimize clustering, providing a compelling baseline for comparison against traditional K-Means applied to the Customer Personality Analysis data.

ANIKA TAZIN PRANTY
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Section 1 — Individual Paper Summaries

Papers_ANIKA TAZIN PRANTY

Paper 1: Retail Industry Analytics: Unraveling Consumer Behavior through RFM Segmentation and Machine Learning

Citation: Arefin, S., Parvez, R., Ahmed, T., Ahsan, M., Sumaiya, F., Jahin, F., & Hasan, M., 2024, *IEEE International Conference on Electro Information Technology (eIT)*

Summary: This paper presents a well-structured retail analytics pipeline that combines RFM segmentation with advanced machine learning classifiers to analyze consumer behavior and predict customer lifetime value. The high accuracy scores confirm that machine learning is highly effective for customer segmentation and targeted marketing decision-making. It is directly relevant to our project as it applies a very similar methodology on real-world retail data. However, its exclusive reliance on transactional data highlights a key opportunity our project addresses — using a richer personality-based dataset for deeper and more meaningful segmentation.

Results: The models delivered strong performance metrics: 92.40% accuracy, 92.27% precision, 92.40% recall, 92.28% F1-score, and 97.39 AUC, with XGBoost and LGBM emerging as the top performers. The RFM framework successfully revealed distinct consumer behavior patterns aligned with seasonal trends and marketing initiatives.

Limitations: The study relies entirely on transactional features such as recency, frequency, and monetary value while ignoring demographic or psychographic attributes. The dataset is limited to UK-based retail customers, which may reduce generalizability to other markets. Additionally, the static segmentation model may not effectively capture evolving customer behavior over time.

Paper 2: A Framework for Customer Segmentation to Improve Marketing Strategies Using Machine Learning

Citation: Ashraf, A., Rayed, C. A., Awad, N. A., & Sabry, H. M., 2025, *Procedia Computer Science — Elsevier*

Summary: This paper provides a clear and practical ML-based framework specifically designed to improve marketing strategies through better customer segmentation. By comparing four clustering algorithms, it offers useful insights into selecting the most suitable method for RFM-based segmentation tasks. The work is closely aligned with our project as it directly connects data-driven segmentation to marketing strategy improvement. While the framework is methodologically sound, the absence of demographic and behavioral attributes presents a gap that our project fills by working with a more comprehensive customer personality dataset, making our approach a natural extension of this study.

Results: K-Means achieved the highest Silhouette Score of 0.432619, indicating the best cluster cohesion and separation among all tested algorithms. The framework successfully grouped customers into actionable segments, enabling more focused and data-driven marketing decisions compared to traditional manual approaches.

Limitations: The evaluation depends solely on the Silhouette Score as an internal validation metric, which may not fully reflect real-world business usefulness of the segments. The paper does not test actual marketing outcomes after applying the proposed segments, leaving practical business impact unvalidated. Furthermore, the use of only RFM features limits the depth of customer profiling, excluding important attributes like personality or purchase motivation.

Paper 3: Enhancing Customer Segmentation Through Factor Analysis of Mixed Data (FAMD)-Based Approach Using K-Means and Hierarchical Clustering Algorithms

Citation: Sattar, U., Ufeli, C. P., Hasan, R., & Mahmood, S., 2025, *Information — MDPI*

Summary: This paper makes a valuable contribution by directly addressing the challenge of clustering mixed-type datasets containing both numerical and categorical variables — a challenge very relevant to our project dataset. The FAMD-based approach preserves important structural relationships that conventional preprocessing methods tend to ignore. The identification of meaningful segments such as seasonal shoppers and high spenders makes this paper highly applicable to our targeted marketing objective. It provides both methodological inspiration and a practical benchmark for handling mixed-feature customer data in segmentation tasks, making it one of the most relevant references for our work.

Results: FAMD reduced the feature space to three principal components capturing 81.46% of total variance, producing cleaner and more interpretable clusters. Agglomerative Clustering slightly outperformed K-Means with a Silhouette Score of 0.52 at k=4, identifying clear segments such as seasonal shoppers, high spenders, and budget-conscious buyers.

Limitations: The dataset of only 3,900 customers is relatively small and may not fully represent diverse real-world populations. The paper does not compare FAMD against other dimensionality reduction techniques such as PCA or t-SNE, limiting evidence for its superiority. The study also does not translate the identified segments into tested, concrete marketing actions.

Paper 4: Intelligent customer segmentation: unveiling consumer patterns with machine learning

Citation: Kumar, N., 2025, *Journal of Umm Al-Qura University for Engineering and Architecture — Springer*

Summary: This paper delivers a well-structured and interpretable segmentation model using Bisecting K-Means combined with RFM analysis on a large-scale retail dataset. The clear behavioral groupings and additional loyalty tiering system provide directly actionable insights for targeted marketing strategy design. It closely aligns with our project goals by demonstrating how unsupervised machine learning can effectively decode consumer patterns. Its limitation of relying only on transactional data actually highlights the added value our project brings by incorporating personality and demographic features for deeper and more personalized segmentation outcomes.

Results: Three behavioral clusters were identified: high-spending frequent buyers, completely inactive users, and recent but low-spending customers. Customers were further ranked into four loyalty tiers Platinum (1,263), Gold (1,324), Silver (982), and Bronze (770) with a Silhouette Score of 0.65 and Davies–Bouldin Index of 0.48 confirming strong cluster quality.

Limitations: The model relies solely on transactional RFM data without incorporating demographic or psychographic attributes, limiting the depth of customer understanding. The study produces only three clusters, which may oversimplify the customer base and miss more nuanced sub-segments. The framework has not been validated across different industries beyond retail, reducing its generalizability.

Paper 5: Analyzing the Dynamics of Customer Behavior: A New Perspective on Personalized Marketing through Counterfactual Analysis

Citation: Ebadi Jalal, M. & Elmaghraby, A., 2024, *Journal of Theoretical and Applied Electronic Commerce Research (JTAER) — MDPI*

Summary: This paper introduces a highly innovative framework that goes well beyond traditional static segmentation by combining dynamic time series clustering with counterfactual analysis, allowing businesses to predict and simulate the effect of specific marketing strategies on individual customers over time. It addresses the gap in existing research where approaches often lack the granularity required for personalized marketing at the individual level and overlook the analysis of customer transitions between different groups. This perspective is deeply relevant to our project as it demonstrates how segmentation can be directly connected to personalized marketing decisions, providing a strong theoretical foundation and future direction for extending our work beyond group-level analysis.

Results: Euclidean distance outperformed all other metrics in distinguishing customer behavior patterns in time series, and Logistic Regression achieved the best prediction accuracy for customer status across segments. The counterfactual analysis successfully enabled decision-makers to simulate and evaluate individualized marketing interventions with high precision.

Limitations: The complexity of the counterfactual analysis framework makes it computationally expensive and difficult to deploy in real-time marketing systems. The study requires high-quality longitudinal transaction data that may not always be accessible for businesses. The framework has also not been tested across diverse industries beyond e-commerce, limiting broader applicability.

SHAH MIRAN APURBO

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Section 1 — Individual Paper Summaries

Customer segmentation is a fundamental strategy in modern marketing, enabling businesses to divide their customer base into distinct groups based on shared characteristics, behaviors, and needs. With the growing availability of large-scale customer data, machine learning methods have become indispensable tools for identifying meaningful segments that support personalized and targeted marketing campaigns. The following literature review examines five key papers that provide theoretical foundations, methodological frameworks, and empirical evidence directly relevant to customer segmentation using the Customer Personality Analysis dataset.

Each paper was sourced from peer-reviewed venues including Springer, MDPI, and ResearchGate, and covers topics including unsupervised clustering (K-Means, DBSCAN), RFM-based feature engineering, systematic reviews of segmentation algorithms, and AI-driven customer profiling pipelines. Together, they provide a comprehensive backdrop for this project.

Paper 1: *Machine Learning-Driven Customer Segmentation and Targeted Marketing: A Systematic Review*

> **Summary:** This systematic review examines 82 research papers published over the past decade that apply statistical, machine learning, and deep learning methods to customer segmentation. The study develops a comprehensive taxonomy of segmentation algorithms — covering K-Means, Decision Trees, Random Forests, XGBoost, DBSCAN, SVM, ANN, and deep learning models such as LSTM and CNN — across industries including retail, banking, and tourism. The authors find that machine learning algorithms, particularly K-Means and ensemble methods, are the dominant tools for segmentation due to their scalability and ability to uncover hidden patterns in large datasets.
>
> **Key Points:** Comprehensive taxonomy of ML algorithms used for customer segmentation; cross-industry application review; discussion of feature engineering importance.
>
> **Key Findings:** K-Means, Decision Trees, and Random Forests are the most widely adopted algorithms. ML-driven segmentation significantly improves targeting precision and marketing ROI compared to traditional statistical approaches.
>
> **Limitations:** The review is limited to papers from a single decade; heavy reliance on public benchmark datasets may not reflect real-world complexity. Ethical and privacy considerations are discussed but not deeply operationalized.

Paper 2: *A Review on Customer Segmentation Methods for Personalized Customer Targeting in E-commerce*

> **Summary:** This Springer paper surveys 105 publications from 2000 to 2022, analyzing customer segmentation methods specifically for e-commerce use cases including retail, banking, and services. The authors propose a four-phase customer targeting process: data collection, customer representation (mainly via manual feature selection or RFM analysis), customer analysis via segmentation, and customer targeting. The survey finds that K-Means remains the dominant algorithm regardless of dataset size or domain, and RFM (Recency, Frequency, Monetary) analysis is the preferred feature representation method.

Key Points: Four-phase segmentation pipeline; comparison of segmentation methods across dataset dimensionalities; temporal analysis of algorithmic trends.

Key Findings: K-Means combined with RFM is recommended as a best-practice approach. No consensus exists on evaluation metrics — 14 different metrics are identified. Deep learning and embeddings remain underexplored for customer representation.

Limitations: The review does not establish a universally agreed-upon benchmark metric. Studies on domain experts' roles in validating segmentation outcomes are largely absent, creating a gap between computational outputs and business decision-making.

Paper 3: *Customer Analysis Using Machine Learning-Based Classification Algorithms for Effective Segmentation Using Recency, Frequency, Monetary, and Time (RFMT)*

Summary: This paper introduces an extended RFMT (Recency, Frequency, Monetary, Time) model for customer segmentation, applied to Pakistan's largest e-commerce dataset. The authors combine K-Means, Gaussian Mixture Models, DBSCAN, and Agglomerative Clustering, using a majority voting technique to determine a stable, consensus-based cluster. Evaluation is performed using multiple internal validity indices, including the Elbow method, Silhouette score, Calinski-Harabasz index, Davies-Bouldin index, and Dunn index. The study also integrates product category segmentation, fiscal year-wise, and seasonality-based segmentation to give marketers a multidimensional view of customer behavior.

Key Points: Novel RFMT feature extension; majority voting ensemble for cluster selection; multi-dimensional seasonal and product-based segmentation.

Key Findings: Majority voting across multiple algorithms yields three stable, meaningful clusters. Adding the Time dimension improves segmentation granularity and enables better customer retention strategies.

Limitations: The dataset is from a single country (Pakistan), limiting geographic generalizability. The study focuses on transactional data only, ignoring demographic and psychographic factors similar to those found in the Customer Personality Analysis dataset.

Paper 4: *How Can Algorithms Help in Segmenting Users and Customers? A Systematic Review and Research Agenda for Algorithmic Customer Segmentation*

Summary: This systematic review examines 172 research articles to comprehensively address algorithmic customer segmentation practices — covering algorithm selection, the number of segments to create, and evaluation methods. The authors identify 46 distinct algorithms used across studies and 14 unique evaluation metrics, highlighting the fragmentation in the field. K-Means clustering emerges as the most frequently applied algorithm. The paper also critiques the tendency of existing work to treat segmentation purely as a computational problem while under-emphasizing its conceptual and business strategy dimensions.

Key Points: 46 algorithms and 14 evaluation metrics catalogued; critique of computational-only framing; research agenda for future work.

Key Findings: Segment size is the dominant hyperparameter used in over 80% of studies. Domain experts are rarely involved in evaluating outcomes, weakening the business relevance of most segmentation research.

Limitations: The review identifies a significant gap: most studies do not integrate qualitative or managerial validation. The field lacks a standardized evaluation framework, making cross-study comparison difficult.

Paper 5: *Customer Profiling, Segmentation, and Sales Prediction Using AI in Direct Marketing*

Summary: This paper proposes a data-driven pipeline for customer profiling, segmentation, and targeted marketing in an Edutech startup context. Applying K-Means clustering with the Elbow method, Silhouette coefficient, and Gap Statistics, the study segments customers based on RFM analysis, identifying three primary clusters: new customers (Cluster A), best customers (Cluster B), and intermittent customers (Cluster C). Each cluster is then matched with a tailored marketing strategy — such as personalized learning content for new users and incentive programs for best customers. Neural networks, fuzzy theory, and evolutionary algorithms are also reviewed as alternative segmentation techniques.

Key Points: End-to-end pipeline from profiling to strategy; RFM-based cluster interpretation; multi-method validation using Elbow, Silhouette, and Gap Statistics.

Key Findings: Three actionable customer segments are identified with distinct behavioral profiles. Cluster-specific marketing strategies improve customer engagement and retention. Gap Statistics validation increases reliability over single-metric approaches.

Limitations: The study is based on a single industry (Edutech) and a relatively small startup dataset, limiting scalability. Demographic variables beyond purchase behavior are not deeply incorporated into the model.

The five reviewed papers collectively establish that machine learning — particularly K-Means clustering combined with RFM-based feature engineering — is the dominant and most effective approach to customer segmentation. Systematic reviews confirm that while 46+ algorithms have been explored, no consensus on evaluation metrics or best practices yet exists. Applied studies demonstrate that multi-cluster pipelines produce actionable marketing strategies, but most are limited by their reliance on transactional data alone. This project builds upon these findings by incorporating demographic and psychographic features from the Customer Personality Analysis dataset, addressing a gap identified across multiple reviewed works.

FINAL — Combined Narrative (All Members)

Section 2 — Combined Narrative Contribution

The question of how businesses understand their customers is not new, but the tools available to answer it have changed dramatically. For decades, companies relied on broad demographic categories — age, income, gender — to divide their customer base and craft marketing messages. This approach, while practical, was blunt. It treated entire populations as monolithic blocks, missing the nuanced behavioral and psychological patterns that actually drive purchasing decisions. The twenty research papers reviewed collectively by this team trace the arc from that early, limited understanding to the sophisticated, data-driven frameworks now reshaping how businesses connect with their customers — and they illuminate precisely where our project stands in that evolution.

The foundational shift began with the widespread adoption of K-Means clustering as the primary engine of customer segmentation. Tabianan et al. (2022) demonstrated that K-Means, applied to e-commerce purchase data and paired with the Elbow method for optimal cluster determination, could effectively identify distinct behavioral profiles — directly linking poor segmentation to customer churn and lost revenue. This established the core pipeline the field would build upon. Kasem et al. (2024) extended this by combining K-Means with RFM (Recency, Frequency, Monetary) analysis and a suite of validation metrics — including the Silhouette coefficient and Gap Statistics — to surface three archetypal customer groups: new, best, and intermittent customers, each demanding a distinct marketing strategy. Yet even as these applied studies demonstrated real-world value, the broader field remained fragmented. Salminen et al. (2023), reviewing 172 studies, found researchers had employed 46 different algorithms and 14 distinct evaluation metrics with no agreed standard — and that domain expertise was rarely involved in judging whether the resulting segments were actually useful for business.

As the field matured, researchers began pushing both the data and the algorithms further. John et al. (2023) challenged the dominance of K-Means with a rigorous five-algorithm comparison on half a million UK retail records, finding that Gaussian Mixture Models outperformed all competitors with a Silhouette Score of 0.80, revealing K-Means' limitations on complex, probabilistic distributions. Ashraf et al. (2025) reinforced this comparative instinct, and Arefin et al. (2024) demonstrated that ensemble models like XGBoost and LightGBM, built on top of RFM-segmented data, could achieve 92.4% accuracy and an AUC of 97.39 — bringing predictive power closer to deployment-ready marketing tools. Wang (2025) pushed further still, proposing K-means-QLDE, a reinforcement learning-enhanced framework that used Q-learning to adaptively tune clustering parameters while employing PCA for dimensionality reduction, a technique that would recur across multiple reviewed studies. The RFMT extension reviewed by Apurbo added a temporal dimension to RFM, and majority voting across multiple algorithms yielded stable, business-relevant clusters that single-method approaches could not match.

Yet a critical gap ran beneath all of this progress: nearly every study worked exclusively with transactional data. Recency, frequency, and monetary value are powerful proxies for behavior, but they say nothing about why a customer behaves that way. Kumar (2025) used Bisecting K-Means to reveal four loyalty tiers — Platinum, Gold, Silver, and Bronze — with a strong Silhouette Score of 0.65, but the model still lacked psychological depth. The truly novel methodological leaps came from two directions. Sattar et al. (2025) addressed the structural challenge of mixed-data clustering head-on, using Factor Analysis of Mixed Data (FAMD) to handle both numerical and categorical variables simultaneously, reducing a rich customer feature space to three principal components capturing 81.46% of total variance — a technique directly applicable to the multi-type Customer Personality Analysis dataset our team is working with. Meanwhile, Ebadi Jalal and Elmaghraby (2024) introduced counterfactual analysis combined with dynamic time-series clustering to simulate individualized marketing interventions, representing the field's most ambitious attempt to move from segment-level insights to truly personalized strategy.

The behavioral and psychological turn in this literature proved equally instructive. Talaat et al. (2023) introduced DeepLimeSeg, fusing deep learning with LIME-based Explainable AI and RFM analysis to produce customer segments that were not only accurate — outperforming K-Means and SVM with an MSE of 0.9412 — but interpretable, directly addressing the 'black box' criticism that has long shadowed AI-driven marketing.

Azad et al. (2023), conducting their research at North South University in Dhaka, took a complementary psychological angle by integrating the Theory of Planned Behavior with ML classifiers, finding that social norms and attitude were the strongest predictors of purchase intent, and that ensemble models significantly outperformed traditional statistical approaches. The 2025 PLOS ONE benchmark confirmed these findings at scale: CatBoost achieved an F1 score of 0.93 and an AUC of 0.985, establishing a clear performance ceiling for our model evaluations to aim toward.

Taken together, the twenty papers reviewed by this team tell a coherent story: the field began with simple behavioral clustering, built toward algorithmic sophistication and ensemble innovation, and is now reaching for something richer — systems that understand not just what customers do, but who they are. The Customer Personality Analysis dataset our project employs is precisely the vehicle for that final step. Where prior studies were constrained by transactional records, our dataset incorporates demographic attributes, lifestyle indicators, product category preferences, and campaign response histories that enable genuine personality-driven segmentation. The methodological toolkit our team will apply — K-Means benchmarked against ensemble alternatives, PCA for dimensionality reduction, and multi-metric validation including Silhouette scores and Davies-Bouldin indices — is directly grounded in the lessons and limitations surfaced across this entire body of literature. This project does not merely replicate what has been done; it directly addresses the gap that the most thoughtful researchers in this field have themselves identified: the move beyond transaction data toward a fuller, more human understanding of the customer.